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Assessment of Landsat-based terricolous macrolichen cover retrieval and change analysis over caribou ranges in northern Canada and Alaska

Blair Kennedy[a,*], Darren Pouliot[a], Micheline Manseau[a], Robert Fraser[b], Jason Duffe[a], Jon Pasher[a], Wenjun Chen[b], Ian Olthof[b]

[a] Landscape Science and Technology Division, Environment and Climate Change Canada, 1125 Colonel By Drive, Ottawa, Ontario K1A 0H3, Canada
[b] Canada Centre for Mapping and Earth Observation, Natural Resources Canada, 560 Rochester Street, Ottawa, Ontario K1S 5K2, Canada

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ABSTRACT

Terricolous macrolichens are an important food source for caribou (*Rangifer tarandus*) and can greatly influence their movement, distribution and demography over time. Mapping the spatial distribution and cover of macrolichens with remote sensing can serve as an important approach for assessing the impact of disturbances (e.g. fire, grazing, trampling) on lichen cover at the landscape scale and for monitoring post-disturbance rates of recovery. Previous remote sensing-based efforts to retrieve the distribution and abundance of lichen have been restricted to particular regions and thus are not indicative of the potential for large extent mapping and monitoring. In this study, we assessed the effectiveness of machine learning methods for retrieving lichen cover and change across different regions in northern Canada and Alaska using Landsat-5 images, topographic and climate data. Global and regional-scale models were evaluated to assess whether regionally specific analyses would improve performance. Of the models tested, the deep neural network was the most accurate for predicting lichen cover (model efficiency (ME) = 0.58, mean absolute error (MAE) < 7%). For the regional analysis, the performance was the best in north-central Canada (ME = 0.56, MAE = 8%) and the worst in north-eastern Canada (ME = 0.22, MAE < 4%) due to lower lichen cover, more exposed ground, and reduced sample quality and distribution. Analysis of trend-based change detection from 1984 to 2011 in the three regional test areas showed the expected directional response with declining lichen cover in north-western Canada in response to climate-induced shrub expansion, slow recovery to wildfire in north-central Canada, and declining lichen cover in north-eastern Canada related to caribou foraging/trampling and shrub expansion.

1. Introduction

Numerous caribou (*Rangifer tarandus*) populations in northern Canada have shown evidence of decline over the last two decades causing concern among various stakeholders, including Indigenous, government, and non-governmental organizations (COSEWIC, 2016; COSEWIC, 2017; Parlee et al., 2018; Vors and Boyce, 2009; Fauchald et al., 2017). Terricolous macrolichens, sometimes referred to as caribou or reindeer forage lichens (mostly in the genus *Cladonia* spp.), are an important winter food source for caribou (Llano, 1956; Jandt et al., 2008; Joly et al., 2010; Joly et al., 2015; Joly & Cameron, 2018). The abundance of forage lichens has been shown to influence caribou habitat use patterns and population demography through time among other complex factors (Crête and Manseau, 1996; Joly et al., 2003; Joly et al., 2010; Jandt et al., 2008). A good understanding of lichen availability is therefore important for the conservation of caribou in northern Canada (Jandt et al., 2008; Rickbeil et al., 2017).

The distribution and abundance of lichens are controlled by physical conditions such as substrate acidity, temperature, moisture, nutrient availability, and sunlight. They are also influenced by natural and anthropogenic disturbances such as wildfire, foraging/trampling by ungulates, climate change, pollution and road construction (Virtanen et al., 2002; Joly et al., 2009; Dahlberg and Bültmann, 2013; Chen et al., 2017). Although the abundance of lichens varies spatially with localized conditions (e.g. moisture, disturbance, competition, etc.), lichen species are generally globally distributed, with as much as 60% of species occurring across all Arctic environments (Dahlberg and Bültmann, 2013).

As the Arctic climate warms (Chapin et al., 2005), there have been corresponding increases to plant biomass (Myneni et al., 1997; Pouliot et al., 2009; Tremblay et al., 2012; Fraser et al., 2014a; Moffat et al., 2016) and in some cases has led to taller and denser canopies,

* Corresponding author.
E-mail address: blair.kennedy3@canada.ca (B. Kennedy).

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conditions that are not favorable to ground lichens due to shading and smothering by accumulating litter. Changes in species composition and canopy structure alter the physical (e.g. thermal regimes, snow depths and melt-patterns) and chemical properties (e.g. soil chemistry through detritus/leaf litter and nutrient uptake) of these systems, thus limiting resources available to support lichen growth (Jandt et al., 2008). Increases in regional temperatures and changes to precipitation patterns have also increased the frequency and extent of wildfires within Arctic and subarctic environments (Joly et al., 2009; Kasischke et al., 2006; Hu et al., 2015; Tsuyuzaki et al., 2018). Lichens are particularly susceptible to fires because they are easily damaged or destroyed even in light wildfires and have been shown to recolonize disturbed sites more slowly compared to vascular plant species (Jandt et al., 2008). In Alaska, forage lichen species recovery can take > 50 years compared to 10 years for graminoids and shrubs (Jandt et al., 2008; Joly et al., 2009). Morneau and Payette (1989) observed the presence of *Cladonia* species 14 years post fire in northern Quebec. However, vegetation cover was not dominated by *Cladonia* species until at least 25 years after fire. It was also shown that *Cladonia* mats gradually thicken until ~130 years post fire. In northern Saskatchewan, Skatter et al. (2014) found similar terrestrial forage lichen species recolonization sequences; however, lichen recovery rates in Jack Pine stands was generally faster than in the east with recolonization occurring ~21–30 years post fire. Lichen abundance is also affected by caribou foraging (including wastage attributed to selective feeding) and trampling particularly when herd sizes are large and environmental conditions are dry. Lichen moisture content is a key factor mitigating mechanical damage by trampling (Crête and Manseau, 1996; Manseau et al., 1996; Jandt et al., 2008; Joly et al., 2009; Heggenes et al., 2017). The recovery of lichens from heavy foraging can take decades as observed in Alaska (Joly et al., 2009); however, the long-term effects of trampling on lichen survivability and reestablishment has not been widely investigated (Heggenes et al., 2018).

The distinct spectral reflectance and absorption characteristics of terricolous macrolichens support the potential for monitoring with optical remote sensing-based technologies in northern boreal and tundra ecosystems (Falldorf et al., 2014; Granlund et al., 2018; Rees, 2004; Petzold and Goward, 1988). Previous remote sensing efforts can be categorized into cover retrieval and classification-based approaches (Falldorf et al., 2014). Retrieval studies seek to predict a continuous estimation of cover or volume, whereas classification defines ranges of cover classes or ecosystem complexes such as lichen-woodland. Nelson et al. (2013) retrieved lichen cover in Denali, Alaska using a non-parametric regression modelling approach. Landsat and elevation data were used to model cover of yellow (high usnic acid), light, and dark coloured lichens. Results showed that yellow lichens were the most accurately retrieved (r^2 = 0.37). Macander et al. (2018) retrieved fractional cover of caribou forage lichens across eastern Alaska and southwestern Yukon using a random forest modelling approach. Training and test data were derived from field plots, unmanned aerial vehicle (UAV) imagery, and airborne surveys. Predictor variables included Landsat seasonal percentile composites and bio-geophysical layers (e.g. elevation, forest cover, and water). Results of their study showed r^2 values ranging from 0.33 to 0.77 based on which dataset(s) were used for testing. In a previous study, Macander et al. (2017) retrieved macrolichen cover on the North Slope of Alaska using random forest, Landsat seasonal spectra and environmental predictor variables with a reported r^2 of 0.55. These studies suggest potential to estimate lichen cover in north-western Canada and Alaska. In northern Quebec, Théau and Duguay (2004a, 2004b) and Théau et al. (2005) applied spectral mixture analysis to extract lichen fractions from Landsat-5 imagery. Results from all three studies showed that fractions identified as containing lichens were correlated with lichen cover.

Lichen volume is a potentially more informative measure than cover of the quantity of lichens available to caribou as forage if it can be retrieved with acceptable accuracy. Falldorf et al. (2014) investigated the use of Landsat for lichen volume retrieval in Scandinavia. Results showed that the developed Lichen Volume Estimator (LVE) can achieve average cross-validated predictions of r^2 = 0.67 within the study area. Rickbeil et al. (2017) applied the LVE in the Northwest Territories to examine its influence on caribou movement; however, lichen volume estimates were not evaluated.

Classification-based approaches have shown overall accuracies in the range of 64–86% when lichen is the spectrally distinct/dominant land cover class, but user's or producer's accuracies can be low (Gilichinsky et al., 2011; Colpaert et al., 2003; Tømmervik et al., 2003; Théau et al., 2005). Classification methods are important for developing land cover characterizations that capture lichen among other class designations. Alternatively, cover estimates can be derived separately and used to enhance existing land cover classifications for specific applications.

Few studies have examined the potential of remote sensing for monitoring lichen cover change. Nordberg and Allard (2002) used Landsat for three change classes with an overall accuracy of 77% for a small study area in Sweden, where change in the normalized difference vegetation index (NDVI) was the predictor. Théau and Duguay (2004b) used fractions from spectral mixture analysis for various bitemporal change comparisons in northern Quebec. No quantitative assessment of the change performance was undertaken, but change maps showed some correspondence with caribou calving ranges. Snow, topography, phenology, spectral and endmember variability were identified as potential sources of error in their analysis. Monitoring lichen distribution and abundance across large (continental) extents can provide greater insight into changing environmental conditions that relate in whole or partly to disturbances. Further, if sufficiently accurate, a remote sensing product can provide information on regional and local variation in post-disturbance recovery rates where ground data are absent. It is therefore of interest to investigate the potential for mapping lichens across large spatiotemporal domains.

1.1. Objectives

The primary objective of this research was to develop and assess a terricolous macrolichen cover retrieval methodology applicable over vast caribou ranges in northern Canada and Alaska that is suitable for change monitoring. Previous studies have derived lichen cover from remote sensing for locally specific geographic ranges, but none to our knowledge have investigated the potential for large regional or national scale analysis across varied landscape conditions. We leverage previously collected field observations from a variety of sources to address the following objectives:

(1) Develop and assess machine learning methods to predict lichen percent cover from Landsat satellite imagery, topographic and long-term climate averages.
(2) Evaluate regional differences in percent cover retrieval and examine the potential for a single global model vs. several regionally specific trained models.
(3) Examine the sensitivity of the retrieval method to known landscape changes in order to assess the potential of the method for monitoring.

2. Methods

2.1. Study sites and lichen sampling methods

Field data representing lichen percent cover were compiled from various sources and locations across Canada and Alaska, referred to hereafter as field sites. Lichen species in these data that exhibited fruticose growth forms and that were generally light-coloured in appearance (mostly belonging to the *Cladonia* genus) were included into the test dataset. Lichen species and associated growth forms outside of this

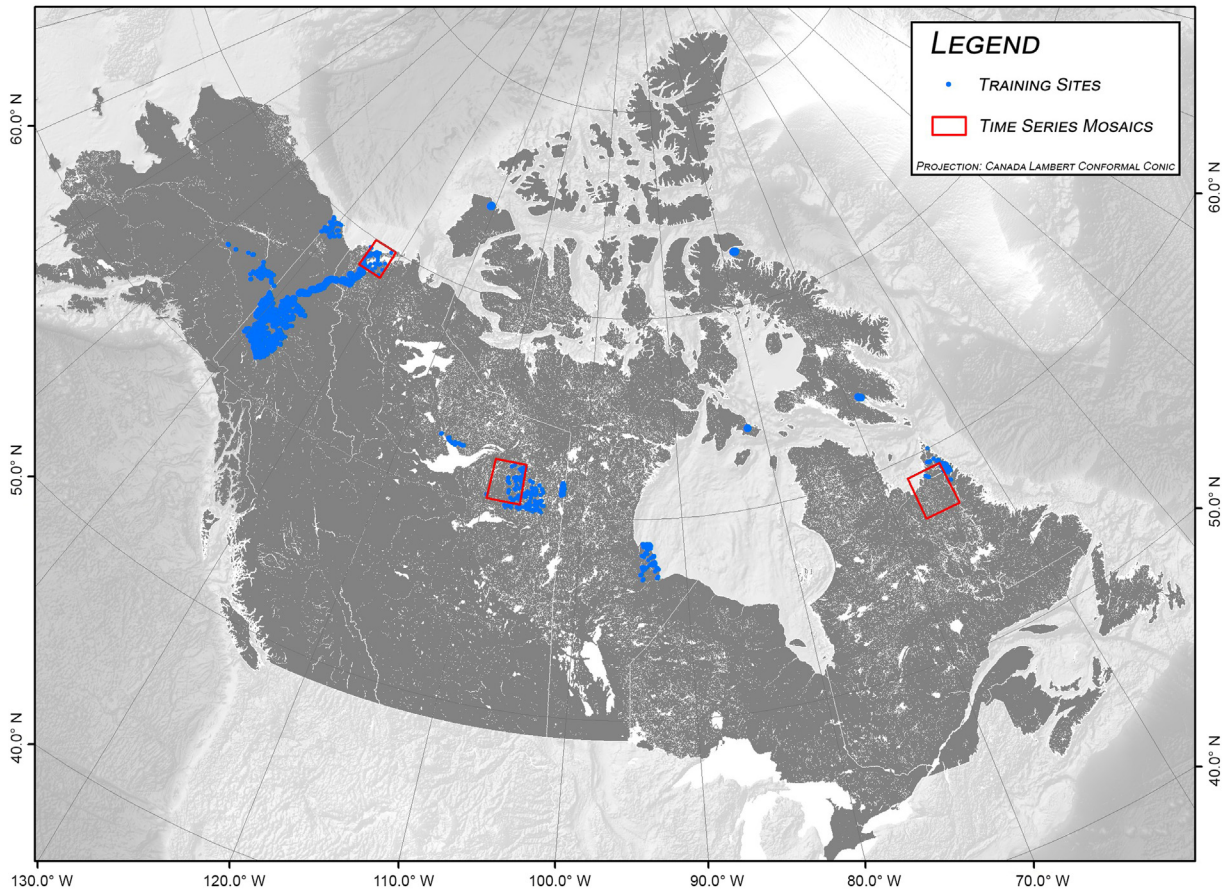


Fig. 1. Map of Canada and Alaska showing ground sampling locations used for model training and testing (lichen percent cover). The red squares are study areas where change detection was assessed referred to as east, central, and west regions. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 1
Field sampling locations, lichen cover statistics, and associated lichen sampling methods.

Region	Year	Samples	Sampling method	Cover			Data source
				Mean	95%ile	Max	
Yukon, Ivvavik National Park	2008	124	Visual from field photos (N, E, S, W)	2	5	12	Parks Canada
Yukon, Dempster Highway	2000	247	Visual from field photos (N, E, S, W)	4	26	45	NRCan/CCRS
Yukon, Tombstone	2010	54	Visual from five 1 × 1 m sample frames	13	45	55	NRCan/CCRS
Yukon/Alaska	2010–2017	5557	Point intercept plots, UAV classification, aerial visual from helicopter	14	45	80	Macander et al. (2018), Yukon Environment
Northwest Territories, Tuktoyaktuk	2018	30	Downward hemispherical photos	26	60	65	ECCC/Landscape Science and Technology
Northwest Territories, Aulavik National Park	2012	39	Downward hemispherical photos	0	1	1	ECCC/Landscape Science and Technology
Northwest Territories, Yellowknife	2005	96	Visual from field photos (N, E, S, W)	3	15	80	NRCan/CCRS
Northwest Territories, South Rae	2013	185	Visual from helicopter aerial photos	8	50	70	NRCan/Geological Survey of Canada
Manitoba, Wapusk National Park	1998/1999	575	Visual in situ assessment	11	60	80	Brook and Kenkel (2002)
Nunavut, Iqaluit	2005	50	Visual from field photos (N, E, S, W)	7	30	70	NRCan/CCRS
Nunavut, South Hampton Island	2015	44	Visual in situ assessment	2	10	10	ECCC/Wildlife Research Division
Nunavut, Pond Inlet	2010	103	Visual from field photos (N, E, S, W)	4	14	18	NRCan/CCRS
Labrador, Torngat Mountains	2008	380	Visual from field photos (N, E, S, W)	5	25	40	Parks Canada

ECCC - Environment and Climate Change Canada, NRCan - Natural Resources Canada, CCRS - Canada Centre for Remote Sensing, cover values are in percent.

description (i.e. crustose lichens appearing on rocks) were treated as non-target species and given a zero percent cover value. These datasets represent a significant investment in data collection particularly regarding the remoteness of each location. Consequently, they provide an effective starting point for assessment. Fig. 1 shows the locations of the field sites used for model training and testing. Table 1 provides details on each of the field data sources included in the analysis.

Lichen cover values in these data were largely based on visual cover estimates made in situ and from downward or oblique landscape photographs (taken in cardinal directions and estimates averaged). All samples were selected to be representative at the moderate spatial resolution (30 m). UAV cover estimates were taken from Macander et al. (2018) in which lichen cover evaluations were based on image classification methods (i.e. supervised classification using a maximum-

Table 2
Summary of the spectral indices (SIs) calculated using Landsat seasonal percentile data.

Spectral index	Acronym	Formula	Reference
Normalized difference vegetation index	NDVI	B4 − B3 / B4 + B3	Rouse et al. (1974)
Normalized difference moisture index	NDMI	B4 − B5 / B4 + B5	Goodwin et al. (2008)
Normalized difference lichen index	NDLI	B5 − B2 / B5 + B2	Nordberg (1998)
Normalized difference water index	NDWI	B2 − B4 / B2 + B4	McFeeters (1996)
Normalized difference snow index	NDSI	B3 − B5 / B3 + B5	Riggs et al. (2015)
Normalized burn ratio	NBR	B4 − B7 / B4 + B7	Key and Benson (1999)

likelihood approach). The lichen cover dataset from the Yukon used in Macander et al. (2018) was derived from aerial-based visual estimates. Hemispherical photos, sampling frames, or point intercept plots were averaged to be representative at the moderate spatial resolution. High resolution satellite imagery in Google Earth and ArcGIS Earth were used to sample non-lichen land cover types that might be spectrally confused with lichen in the visible wavelengths (e.g. Landsat 5 bands 1–3). Cover types, such as developed areas, sandy shorelines, riverbed exposures, water with high sediment loads, snow/ice/aufeis, and rock outcrops were identified as suitable non-lichen training samples. Between 10 and 20 samples of these cover types were added to most of the field sites.

Subjectivity in visual estimates, geolocation, and the position of the sample relative to the Landsat grid are sources of error in the data. Ideally, all data would be collected following a single field protocol coincident with image acquisitions, but cost and utilization of existing data were the main considerations in this study. To reduce errors, for each site listed in Table 1 we applied multiple linear regression using the Landsat, topographic, and climate predictors to identify potential outliers in lichen cover. Outliers were identified through residual analysis and subsequently checked to determine the source of error (e.g. interpretation error of field data or artifacts in the Landsat imagery). In many cases outliers were related to disturbance events, such as wildfires, that had altered the original field-based lichen estimation or that occurred after field data collection but before Landsat imagery was acquired. For some sites, interpretation was adjusted to remove crustose lichens which were not considered a significant source of forage for caribou. Residual cloud or shadow contamination in the Landsat data were identified in a few cases and eliminated from the dataset – such errors were attributable to the performance of Fmask algorithm (Qiu et al., 2019). Outliers that were not related to an identifiable error were not removed or adjusted otherwise.

In all there were 13 different field sites with a total of 7484 samples (Table 1). Most of the data were collected in north-western regions; with fewer data available in north-eastern Canada. Maximum lichen cover values were high at all locations (e.g. 70–80% cover) except for sites in the high Arctic (e.g. Aulavik National Park and northern Baffin Island, 0–10% cover) and for most sites in the north-east. The occurrence of high lichen cover (> 50%) was the most frequent for sites in north-central Canada.

2.2. Landsat and geophysical data

Landsat 5 TM Collection 1 Tier 1 calibrated top-of-atmosphere (TOA) reflectance data was used to facilitate the assessment of lichen cover change from 1984 to 2011. Low solar elevation angles combined with reduced quantities of dense dark vegetation (used for the estimation of aerosols) make surface reflectance more uncertain at high latitudes. For these two reasons, the United States Geologic Survey (USGS) Landsat product guide warns against the use of surface reflectance data products for latitudes above 65° (United States Geologic Survey, 2019). Therefore, to develop methods that were applicable across the three study sites, we did not use surface reflectance. Landsat 5 TOA data were composited using percentiles for the desired time periods (e.g. seasonally and yearly) using Google Earth Engine. Cloud and cloud shadow flags from the data quality layer were first applied and the percentile estimates generated using three years of data to minimize residual contamination (e.g. clouds, shadows, and haze). These data were synchronized to within ± 1 year of field observations; compositing over three years was necessary to produce clear observations of each site. Data were exported as both sample points and image composites. Image composites were reprojected to UTM coordinates for each site.

Four approaches were used to define Landsat seasonal feature sets. These included: (1) summer only (SUM); (2) spring, summer, fall (SPR-SUM-FALL); (3) annual percentiles of 20%, 40%, 60%, 80% (P20-40-60-80); and (4) a reduced set of annual percentiles of 20%, 50%, 80% (P20-50-80). For all feature sets three years of data were pooled to reduce residual cloud, cloud shadow, and haze. The annual percentiles were computed using all cloud free reflectance observations from June to September. Interpretation of the annual percentile composites must be considered for a specific band. For example, the 80th percentile (brighter values) in the red band represents a combination of leaf-off and drier ground conditions across years, whereas the lower 20th percentile (darker values) represents wetter and more vegetated conditions. The annual percentiles had reduced cloud, cloud shadow and missing data relative to the SPR-SUM-FALL feature sets and thus were included in the analysis. Annual percentiles were previously used in Macander et al. (2018). The 40th percentile was used for the SUM and SPR-SUM-FALL image composites. This criterion was selected to reduce clouds and haze which tend to increase reflectance in most bands and tend to be more frequent than shadow contamination. Various spectral indices (SIs) were derived from these datasets (Table 2) and included into each of their respective seasonal feature sets.

Elevation, aspect, and slope were used as surface terrain predictors to represent habitat suitability for lichen. The Arctic Digital Elevation Model (ArcticDEM, Porter et al., 2018) was downloaded at 10 m spatial resolution resampled to 30 m using bilinear interpolation and then reprojected to UTM for each of the study areas. Bilinear resampling was used to account for grid offsets and geolocation errors between Landsat and terrain variables. For sites outside of the ArcticDEM coverage (two sites below 60° latitude), the Canadian Digital Elevation Model (CDEM, Natural Resources Canada) was used. These data were downloaded, converted to ellipsoidal height from orthometric heights, and reprojected to the desired UTM zone. Aspect and slope were subsequently calculated from the resampled and reprojected datasets. Examination of these datasets revealed similar structure and correlation at the 30 m spatial resolution. For use in linear models, aspect data was transformed into sine and cosine components. For the non-linear models, aspect was used directly.

Long-term temperature and precipitation gridded data were included to improve environmental characterization of potential lichen habitat (i.e. data were used as a regional ecological stratifier). These data were acquired from the WorldClim Global Climate Data Version 2.0 spatially interpolated and gridded 1970–2000 dataset at a resolution of 30″ (Fick and Hijmans, 2017). The bioclimatic data layers used included: (a) mean temperature of the warmest quarter (BIO10), and (b) mean precipitation of the warmest quarter (BIO18). Mean temperature of the warmest quarter is similar to the “summer warmth index” (Walker et al., 2003; Epstein et al., 2008). Walker et al. (2003) have previously demonstrated in Alaska that (a) the summer warmth

index is positively correlated with NDVI, and (b) total plant phytomass, leaf area index, and max NDVI increases with summer temperatures along climatic gradients associated with the summer warmth index. Mean precipitation of the warmest quarter was selected as it provides an approximation of the amount of moisture that would be available within a region. Sample points were extracted using the nearest neighbour from these locations for use in modelling.

2.3. Lichen cover modelling

2.3.1. Models

Data exploration revealed non-linear relationships between lichen cover and input features (i.e. predictors). The non-linear methods tested included random forest (RF) and a deep neural network (NN). For baseline comparison two linear methods were also evaluated, multivariate least squares regression (MLR) and the least absolute shrinkage (LASSO, Tibshirani, 1996). LASSO was removed after initial testing did not reveal any advantage relative to MLR. All methods were implemented in Python using the Scikit-learn and Tensorflow packages.

To tune parameters in RF, a randomized search strategy was applied followed by grid search refinement. For the randomized search, we used the number of estimators (trees) ranging from 100 to 1000, number of features considered at each split (square root of total features), minimum samples for splitting (2–6), and minimum samples per leaf (1–5). For a description of these parameters see the Scikit-learn documentation for RF (Scikit-learn, 2018). We generated 500 randomized parameter sets and used 3-fold cross-validation to identify the optimal set. For the follow-on grid-based refinement step, we used the results from the randomized search and set the parameter ranges to ± 50 for the number of estimators and ± 1 for the other parameters except for the number of features where ± 3 features were tested. We examined using several of the grid-based refinement iterations (3–10) but did not find consistent improvement relative to a single grid search step. Multiple iterations of the grid-based refinement often resulted in reduced accuracy related to model overfitting.

The NN included 7 hidden layers plus input and output layers (Fig. 2). Residual connections were applied to enhance learning, which has been shown to be effective in He et al. (2016). Batch normalization was applied after all hidden layers to improve model generalization (Loffe and Szegedy, 2015). A simple data augmentation strategy was also used for this purpose where a random ± 3% adjustment of the input value for 80% of the inputs was applied. Activations were hyperbolic tangent as they were found to improve performance in initial testing over regularized units. The activation for the last output layer used a sigmoid activation to generate output in the range of 0 to 1.

For training the NN, the Adam optimizer was used with 500 epochs and mean squared error loss function. Batch size was set to 32. To determine training convergence, we generated a 20% hold-out sample from the training data. Improvement with the hold-out was monitored after each epoch and training stopped if no improvement was found in 50 epochs. This was iterated 5 times and results average across the five models for comparison with independent test data. All NN were trained using a Tesla M100 GPU. For a description of deep leaning neural networks see Nielsen (2015) and Ruder (2016) for a comparison of network optimization methods.

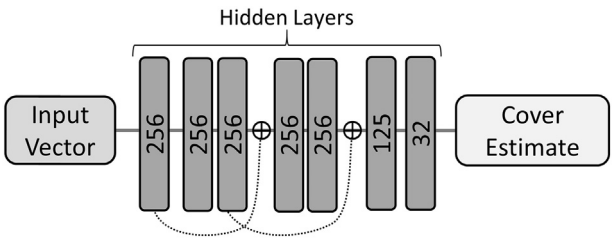


Fig. 2. Schematic of the neural network structure used in the study. Numbers for the hidden layers indicate the number of nodes and lines show layer connectivity. The ⊕ sign represents the addition of layer outputs.

2.3.2. Model evaluation

Several metrics were used for evaluation: model efficiency (ME; Nash and Sutcliffe, 1970; Mayer and Butler, 1993), coefficient of determination (r^2), root mean squared error (RMSE), mean absolute error (MAE), and 5th and 95th error percentiles. These metrics have been reported in other studies and thus this set was included to facilitate comparison. The ME is essentially the r^2 where the sum of squared errors is computed between the model prediction and test data without recalibration. ME can range from -infinity to 1. Negative values occur when the observed mean is a better predictor than the model. The r^2 is not an indication of independent model performance if it incorporates a recalibration to test data (Mayer and Butler, 1993). It was included to provide context for interpreting the ME relative to other studies and identify the potential model improvement with recalibration.

2.3.3. Sampling

To develop a robust model, we applied a resampling scheme to create training and test data for seven iterations. For each iteration, we applied a 600 m distance threshold to reduce spatial autocorrelation between the training and test samples. For points selected as the initial training sample, a 600 m buffer was applied and points falling within the buffer were added to the training set. This process was iterated until no further points were added to the training dataset. For each iteration we sought a 20% hold out test sample, but due to the autocorrelation criteria this varied slightly across iterations in the range of ~1%.

The 600 m distance was selected based on initial models runs using RF with different distance thresholds. This showed that the influence of spatial autocorrelation on the ME metric dropped rapidly from 5 m to 200 m, after which the effect was much less evident. Another factor taken into consideration was the inter-neighbour sample distance for each field site. The majority of samples for the field sites were within a distance of < 600 m to a neighbouring sample. Thus, using a distance > 600 m excluded a large number of samples from the test sites where sample neighbour distances were small.

To improve the probability of selecting samples in each iteration across field sites, we upsampled and downsampled the data for each site to have a minimum of 500 and maximum of 1500 samples. However, we did not allow the upsampling to exceed a factor of 3. The Alaska-Yukon dataset (Macander et al., 2018) was large and to avoid it strongly biasing the results, we downsampled it for each iteration from the full set of observations ($n = 5557$) to 1500 for training and 200 for testing. The full dataset and realized training and test sample distributions (mean and standard deviation across the seven iterations) by lichen cover and region (west, central, east) are shown in Fig. 3. The difference between the full dataset and training sample in the east is the result of upsampling to reduce regional biases in the results. For each region, the sampled test distribution differed from ~500 in the west to ~120 in the east for each iteration. To reduce the effect or regional imbalances, we also computed the above model evaluation metrics for an adjusted test sample, where the test data was unsampled across sites to be the same.

2.3.4. Seasonal feature sets and feature selection

We applied all models to the seasonal Landsat data and selected the feature set with the best performance. To investigate the potential of further reducing the seasonal input features we applied a permutation importance change measure following Breiman (2001). Each input variable was permutated keeping all other inputs unchanged, the model was then run to examine variable effect on the ME. The permutation breaks the correlation with the target variable; variables that cause the largest decrease in ME have the greatest importance in the specific model. Based on this analysis, we evaluated feature subsets from 10 to 50, with a step of 5.

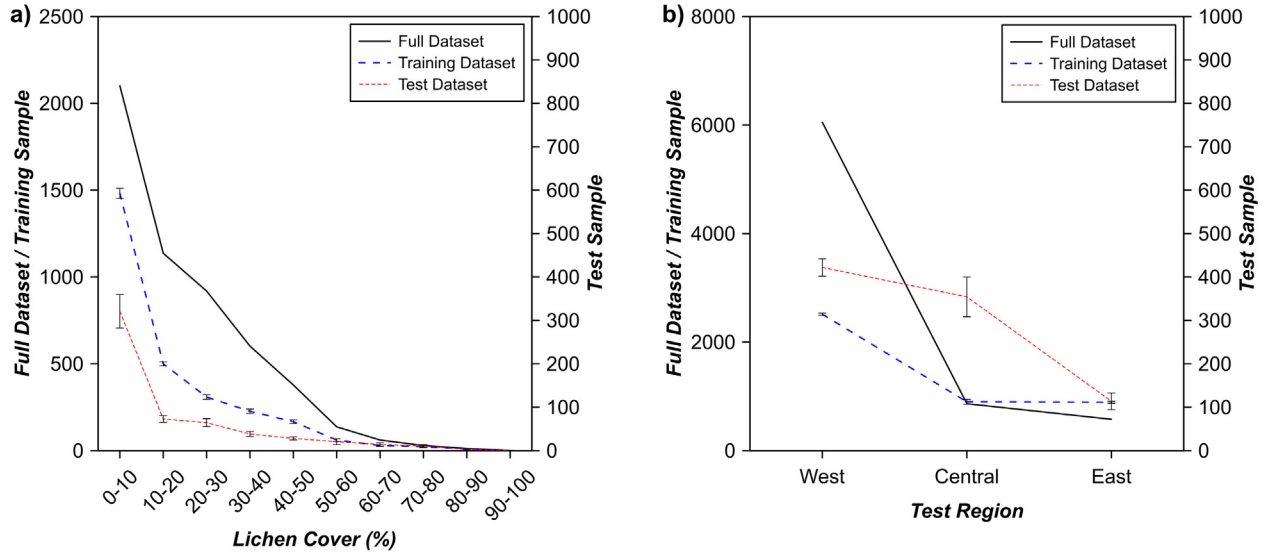


Fig. 3. Full dataset, training and test sample distributions by cover (a) and region (b). Training and test distributions are the average from the seven resampling iterations. Error bars represent ± 1 standard deviation. To improve visualization, the lichen cover (top) in the range 0–10 was divided by a factor of 2 for the full dataset, training, and test samples.

2.3.5. Comparison of global and regionally trained models

A global model was compared to regional models with the objectives of (1) determining if regional training provided improvements over global training and (2) identifying regional differences in model performance. The global model was trained using data from all test sites/regions, whereas the regional models were trained using regionally specific data. The same resampling approach was applied using seven iterations, where in each iteration global training, regional training, and regional test sets were developed. Input features were the best set determined from the feature selection analysis.

2.4. Assessment of time series change potential

The ability to track change over time is a desirable attribute and is ultimately the motivation for using Landsat to estimate lichen cover. To assess this potential, we applied our final model in regions with known landscape changes (Fig. 1, red polygons). Within these study regions we examined areas of no change, sources of noise (e.g. within imagery), and whether there was evidence supporting the detection of the change event. If the model is highly sensitive to noise and has limited correspondence to known change events, then it is clearly not suitable for monitoring. We used Landsat time series for 7 time periods of 1984–1987, 1988–1991, 1992–1995, 1996–1999, 2000–2003, 2004–2007, 2008–2011 to estimate lichen cover. For each period we computed the mean cover from the seven model training iterations and detected change for the time series using the Theil-Sen trend estimator (Sen, 1968) and Mann-Kendall trend significance (Kendall, 1975). The lichen cover time series was then assessed in three large regions (~20,000 to 45,000 km^2) of northern Canada (north-east, north-central, and north-west) representing different drivers of landscape change.

2.4.1. North-east

In the north-east region, caribou foraging and trampling has been observed to reduce lichen cover (Manseau et al., 1996). Climate change and related shrub expansion (greening) has also been identified as a driver of ecosystem change (Myneni et al., 1997; Pouliot et al., 2009; Tremblay et al., 2012; Ju & Masek, 2016; Allard and Lemay, 2012; Campeau et al., 2019). We did not examine the potential interaction effects of caribou foraging and climate change on lichen decline as it was considered beyond the scope of the change assessment objective. Here we compared our lichen change predictions with the field observations of Manseau et al. (1996) in order to assess the agreement of detected changes.

2.4.2. North-central

In the north-central region, fires were the dominant driver of ecosystem change impacting more than ~50% of the area over the period examined. For example, there were a total of 238 fires with a mean size of ~78 km^2 between the years 1984 and 2011. To examine the response and recovery of lichens to fires, we extracted the mean predicted lichen cover for pixels by fire year based the National Fire Database polygons (Canadian Forest Service, 2019). The polygons were rasterized into the 30 m resolution and projected to the appropriate UTM zone. Lichen cover estimates for each composite period (e.g. 1984–1987) were extracted and plotted in order to examine the change timing and recovery rates.

2.4.3. North-west

In the north-west region, we used data from Fraser et al., 2014b in which lichen cover change was estimated and compared for the years 1980 and 2013 using high-resolution (3–4 cm resolution) air photo pairs. Photo dimensions ranged in size from 40 to 90 m. Thus, to extract values for comparison with these data, we digitized the footprint of each photo and used it to extract the Landsat-based lichen predictions. The Landsat predictions were resampled to 2 m to extract only the portion of the photo covering a 30 m pixel. The photo dimensions are approximately the same as a Landsat pixel, but due to grid alignment can intersect several Landsat pixels. This is problematic as lichen tends to be spatially heterogenous and thus the lichen contribution for the overlap between footprints cannot easily be factored out from the non-overlap footprint area of a Landsat pixel. Another challenge with this comparison was the temporal consistency between these data. To reduce this effect, we used the trend estimate to linearly extrapolate our cover estimates from the 1984–1987 composite to 1980 and the 2008–2011 composite to 2013. To account for model bias and sources of error between these data we elected to compare relative change. For relative change, we estimated the 1984 and 2011 lichen cover based on the Theil-Sen fit, computed the difference, and divided by the 1984 estimate.

Table 3
Comparison of feature sets and model performance metrics using the independent test samples across the 7 iterations. Feature sets are described in Section 2.2. Landsat and geophysical data.

Feature set	Model	ME	r^2	RMSE	MAE	P5%ile	P95%ile	ME_{adj}	MAE_{adj}	$RMSE_{adj}$
P20-40-60-80	NN	0.585	0.595	10.447	6.095	−9.882	23.328	0.583	5.459	9.572
		0.016	*0.016*	*0.532*	*0.243*	*1.069*	*2.496*	*0.018*	*0.258*	*0.542*
P20-40-60-80	RF	0.537	0.561	10.983	6.719	−11.750	24.606	0.534	5.999	10.140
		0.026	*0.032*	*0.605*	*0.339*	*0.655*	*2.855*	*0.022*	*0.409*	*0.681*
P20-40-60-80	MLR	0.258	0.264	13.915	9.855	−14.754	32.535	0.223	9.237	12.994
		0.013	*0.009*	*0.635*	*0.304*	*0.761*	*3.141*	*0.036*	*0.353*	*0.692*
P20-50-80	NN	0.564	0.578	10.670	6.164	−10.508	24.179	0.564	5.525	9.878
		0.032	*0.022*	*0.771*	*0.312*	*1.258*	*2.956*	*0.033*	*0.333*	*0.687*
P20-50-80	RF	0.536	0.563	11.003	6.806	−11.843	24.221	0.529	6.105	10.278
		0.029	*0.034*	*0.659*	*0.334*	*0.606*	*3.073*	*0.024*	*0.400*	*0.607*
P20-50-80	MLR	0.262	0.268	13.881	9.741	−14.557	32.897	0.233	9.087	12.959
		0.014	*0.010*	*0.665*	*0.327*	*0.729*	*3.402*	*0.032*	*0.347*	*0.684*
SPR-SUM-FALL	NN	0.585	0.596	10.400	6.059	−11.505	22.167	0.581	5.468	9.633
		0.021	*0.021*	*0.456*	*0.214*	*1.599*	*2.158*	*0.016*	*0.244*	*0.493*
SPR-SUM-FALL	RF	0.497	0.523	11.448	6.846	−11.971	25.389	0.487	6.164	10.619
		0.036	*0.043*	*0.642*	*0.301*	*0.797*	*3.168*	*0.033*	*0.351*	*0.715*
SPR-SUM-FALL	MLR	0.244	0.252	14.040	9.651	−14.355	32.339	0.209	9.126	13.255
		0.016	*0.015*	*0.603*	*0.324*	*0.685*	*3.062*	*0.021*	*0.293*	*0.670*
SUM	NN	0.478	0.486	11.657	7.005	−13.456	25.761	0.483	6.228	10.659
		0.034	*0.031*	*0.471*	*0.226*	*1.578*	*2.967*	*0.032*	*0.302*	*0.600*
SUM	RF	0.479	0.506	11.651	7.256	−12.347	26.931	0.476	6.465	10.685
		0.033	*0.032*	*0.666*	*0.331*	*0.936*	*3.381*	*0.032*	*0.381*	*0.670*
SUM	MLR	0.166	0.174	14.745	10.105	−13.647	34.118	0.161	9.170	13.574
		0.016	*0.017*	*0.608*	*0.325*	*0.612*	*2.774*	*0.017*	*0.308*	*0.670*

ME - model efficiency, r^2 - r squared, RMSE - root mean square error, MAE - mean absolute error, P5%ile - 5th percentile of residual error, P95%ile -95th percentile of residual error, adj - balanced sample adjusted estimates.
Numbers in italics are the standard deviation across resampling iterations.

3. Results

3.1. Seasonal feature set comparison

Comparison of the models across features sets showed that the NN preformed the best (ME = 0.48 to 0.58), followed closely by RF (0.48 to 0.54; Table 3). The linear model did not achieve a reasonable result (ME = 0.17 to 0.26) fitting the low (< 20%) and disregarding the higher cover values. The full season feature sets (SPR-SUM-FALL, P20-40-60-80, P20-50-80) all preformed similarly, to within ± 1 standard deviation. The summer (SUM) only feature set did not preform as well, producing the lowest scores across all metrics for a given model. The regionally adjusted assessment metrics showed that the ME was slightly reduced in most cases and the MAE decreased. The MAE decreased because the proportion of test data in the eastern region was greater, which due to the reduced data range, has a smaller MAE compared to the other regions. RF was more sensitive to the sample adjustment compared to the NN and the linear model was the most sensitive. The NN was best able to account for regional differences and supports the potential for modelling across regions with a single model. Based on these results, we elected to use the P20-40-60-80 feature set because missing data, residual clouds and shadows were reduced relative the SPR-SUM-FALL features. It also showed slightly better performance than the reduced-percentile seasonal feature set (P20-50-80) and the increased number of features was considered an advantage in developing an optimal predictor subset.

Fig. 4 shows scatterplots for three of the resampling iterations and a kernel density plot containing results from all seven iterations. These show a general bias toward underestimation of larger values > 20%. This is the result of the model attempting to balance overestimation of low lichen cover (< 20%) by underestimating larger cover values. Upon closer inspection it appears that low and high lichen cover can have similar spectral properties. For example, high lichen vs. high rock/sand covered areas or high understory lichen vs. vegetation cover without lichen show similar spectra, respectively. There is also a considerable spread of errors, which is in part due to the quality of field data and errors in the composited Landsat data. Averaging across the seven iterations, can reduce this error component as indicated by the density plot, which shows better correspondence of high-density predictions along the 1 to 1 line. Across the seven iterations the average multiplicative recalibration factor was 1.1.

3.1.1. Feature selection

The permutation importance ranking for the NN is shown in Fig. 5. It reveals that the elevation data was the most important variable in the model followed by the environmental variables (temperature and precipitation) and then elevation derivatives (slope and aspect) which were all ranked in the top eight. These variables are considered a measure of lichen habitat in the model. The shortwave infrared band and near infrared bands were the highest-ranking spectral features across the set of seasonal composite percentiles. Visible bands were generally low ranking, except when combined with SIs. The thermal bands were all ranked low. In comparison to the results for RF using the same permutation importance evaluation, the features were generally similar, except that some of the thermal and visible bands were ranked in the top ten. Similar to the NN, the elevation data was ranked as the most important feature in RF.

Based on the feature importance ranking we tested feature subsets from 10 to 50 and examined the change in the ME across seven iterations (Fig. 6). This showed that the ME increased from 10 to 35 features and then leveled-off. Based on these results, we selected the 45-feature set because it had lower variability across the seven iterations and its mean ME was the closest to full feature set.

3.2. Comparison of global and regionally trained models

Building NN models trained in each region did not improve results over the globally trained NN (Table 4). This suggests that the global NN was able to effectively represent regional differences without a substantial trade-off in performance. The larger sample in this case improved model training. However, this is model specific. Examination of regionally adjusted ME in Table 3 revealed that the linear model had

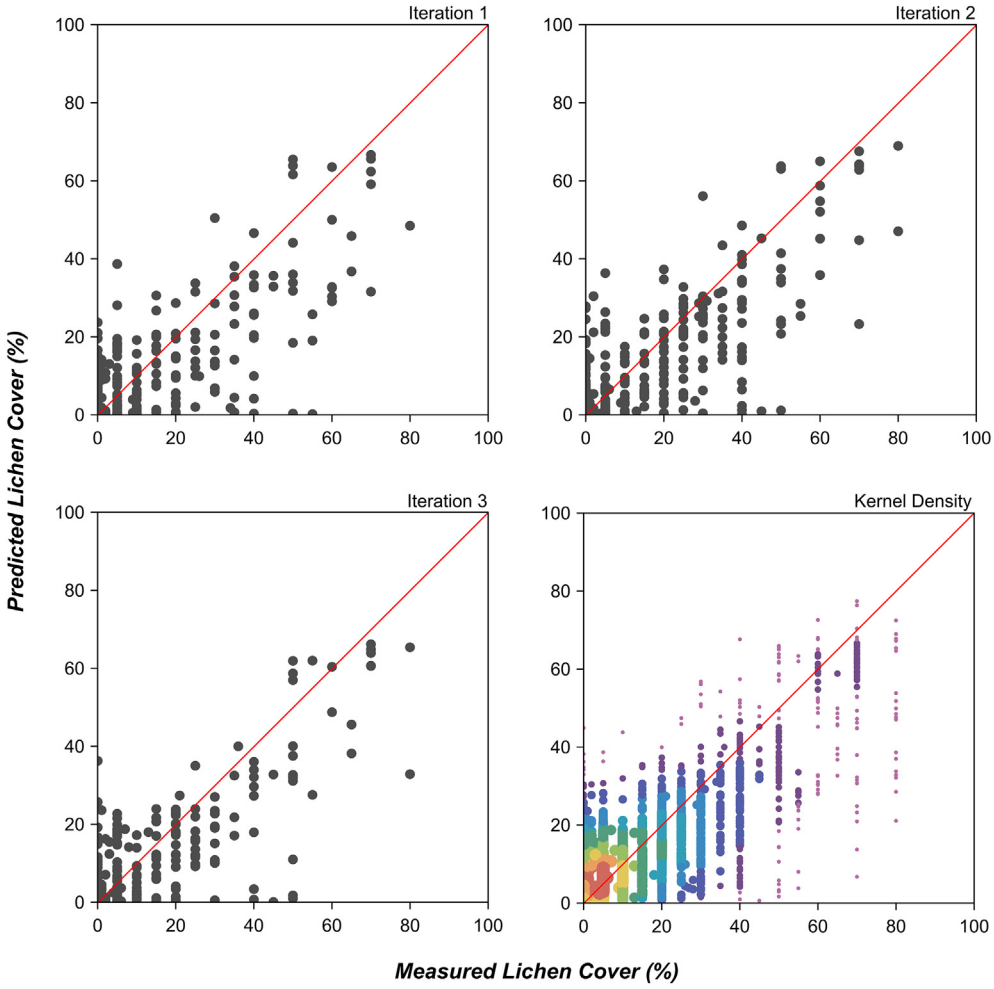


Fig. 4. Scatter plots showing lichen prediction vs. measured values for three iterations using the neural network. Point density is shown (bottom right) for all iterations using varying size and colour of points (i.e. red and larger points demonstrate where density is the greatest). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

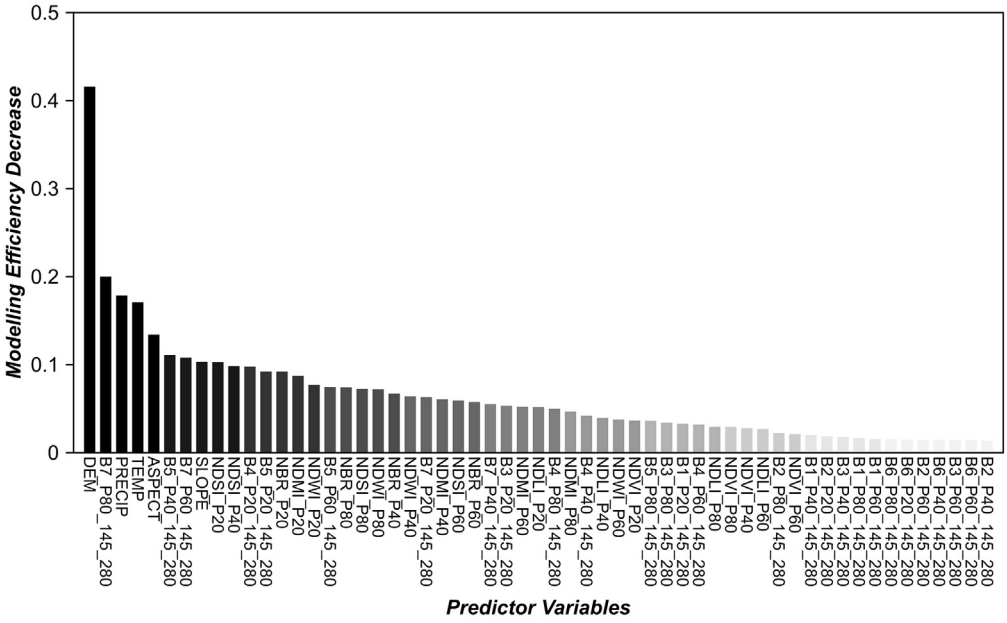


Fig. 5. Decrease in modelling efficiency relative to the full model (ME = 0.78) with variable permutation for the neural network.

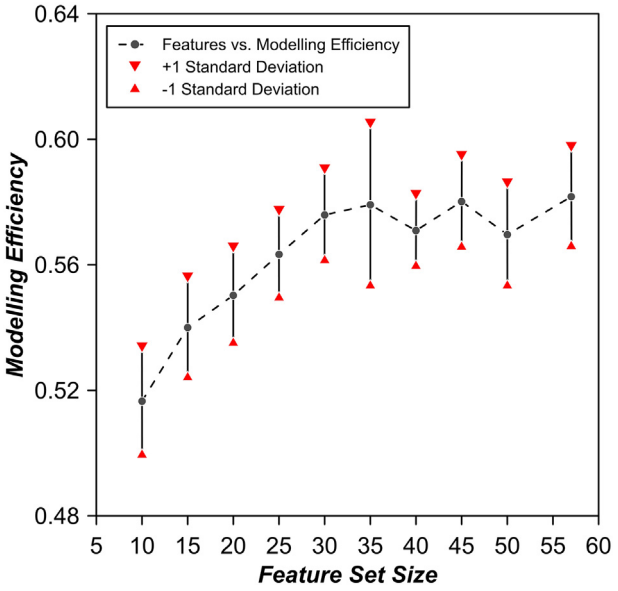


Fig. 6. Performance of feature sets sizes based on the ME decrease ranking. The last point represents the full feature set.

Table 4
Comparison of model performance with regional vs. global training for test regions across the seven resampling iterations.

Region	Training	ME	r^2	RMSE	MAE	5th%ile	95th%ile
East	Regional	0.182	0.235	6.297	4.007	−6.408	14.614
		0.097	*0.071*	*1.315*	*0.405*	*1.792*	*3.974*
East	Global	0.222	0.270	5.919	3.869	−6.530	13.923
		0.110	*0.084*	*0.985*	*0.479*	*1.394*	*3.373*
Central	Regional	0.513	0.554	13.2479	7.942	−11.489	30.119
		0.053	*0.068*	*1.663*	*0.652*	*2.509*	*7.699*
Central	Global	0.557	0.583	12.409	7.068	−10.725	28.360
		0.076	*0.075*	*1.749*	*1.038*	*1.308*	*7.691*
West	Regional	0.533	0.536	9.397	6.234	−12.919	18.388
		0.025	*0.023*	*0.565*	*0.362*	*0.829*	*2.540*
West	Global	0.544	0.547	9.276	6.114	−12.790	19.151
		0.031	*0.031*	*0.451*	*0.260*	*0.787*	*2.515*

ME - model efficiency, r2 - r squared, RMSE - root mean square error, MAE - mean absolute error, P5%ile - 5th percentile of residual error, P95%ile - 95th percentile of residual error.
Numbers in italics are the standard deviation across resampling iterations.

reduced capacity to account for regional variation, whereas the NN demonstrated the greatest potential (largest difference between ME and ME_{adj} in Table 3).

Both the north-central and north-west regions showed similar performance, similar to the best results for the global analysis. The north-east region had lower ME and r^2 measures due to lower lichen cover (e.g. Bland and Altman, 2011). The MAE, RMSE, and error percentiles were the lowest in the north-east due to the reduced range and smaller cover values.

3.3. Assessment of time series change potential

3.3.1. North-east

The results of trend-based change detection are shown in Fig. 7. The greatest negative change in lichen cover corresponds in part to the region identified as being impacted by caribou in Manseau et al. (1996). Examination of the lichen cover maps and trend results revealed errors related to residual clouds, haze, and shadows. There was also variability between years within the expected error range as determined from the modelling analysis of ± 10%. Although infrequent, there were also several pre-1980 fires that showed some increases in lichen mostly in the range of 5–10%, but as high as 30% in some cases for smaller patches. Strong haze occurring over areas with low vegetation combined with the spectral characteristics of lichen caused an increase in lichen predictions in the order of 10–20%. This was only problematic when it occurred over more than one year in the time series which was rare. Otherwise the trend processing was robust. As a point of caution, there were few samples within the bounds of the mapped region and thus greater uncertainty in results is expected. Further, due to the confounding factor of climate change and shrub expansion in the region the detected changes may not be the direct result of caribou foraging related disturbance.

3.3.2. North-central

The mean lichen response for fires occurring in different years is shown in Fig. 8. In all cases fires strongly reduced the lichen cover as expected. Pre-fire variability in lichen cover can be seen for the more recent fires and corresponds to model the error of ~10% and residual contamination in the Landsat data. The more interesting aspect of these results are that post fire lichen cover is not increasing, consistent with published recovery rates.

3.3.3. North-west

Comparison of the model predictions to the cover estimates in Fraser et al., 2014b showed good agreement considering the spatial and temporal sources of error between these data (Fig. 9). Following from the model assessment, a bias toward underestimating larger lichen cover percentages was observed. Comparison of the relative change over time revealed good agreement with almost all estimates showing a reduction in lichen cover. There were however several outliers, two of which were caused by cloud and haze contamination across multiple time periods for the time series in these areas. The other two were due to the use of the Thiel-Sen trend estimator, which is considered robust to 30% outliers in the data. This reduced the sensitivity to valid lichen predictions in some cases resulting in a change magnitude estimate of ~12%.

4. Discussion

4.1. Cover model assessment

Few studies have examined the potential of retrieving lichen cover from moderate resolution sensors such as Landsat, while none have attempted to retrieve lichen cover across such a broad geographical scale over time. Compared to other studies, our model fit, r^2 value of 0.6 and RMSE of 10.4%, was similar to or better than previous analyses. Considering our data sources and the spatial extent examined, this is an encouraging result given that retrieving lichen cover with Landsat is challenging. Recent analysis by Macander et al. (2018), covering the interior of Alaska and part of the Yukon, emphasized the importance of model training and testing when estimating lichen cover. Their best results were achieved with UAV and visual aerial surveys with r^2 ranging from 0.71 to 0.77 and RMSE values ranging from 6.5 to 9.2%. Lower results were obtained with plot surveyed data, with r^2 ranging between 0.33 and 0.70 and RMSE values ranging between 7.7 and 16%. For the test set considered to the most representative an r^2 of 0.69 and RMSE of 8.2% was reported. Differences in sampling, cover estimation method (e.g. plot vs. UAV), and criteria used to account for spatial autocorrelation are likely important factors in the differences between studies.

Earlier remote sensing studies have shown a capacity to quantitatively assess lichen cover using image classification techniques. For example, Gilichinsky et al. (2011) reported class accuracies (producers and users) ranging from 53% to 87%, whereas Théau et al. (2005) reported accuracies of 30% to 92% using more complex classification methods. The results of this analysis are consistent with Gilichinsky et al. (2011) where thresholding cover estimates produced class

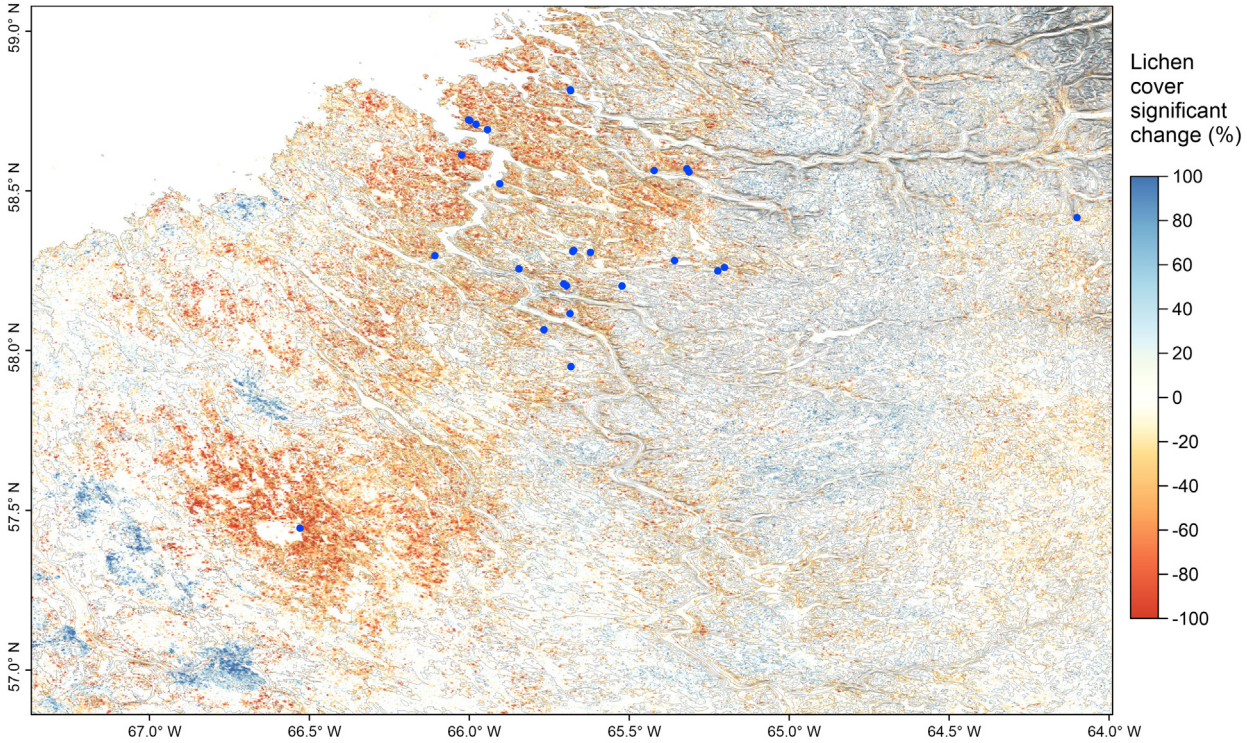


Fig. 7. Significant trends in the east region with sample points of caribou foraging overlaid as blue dots from Manseau et al. (1996). Yellow to red indicates significant decreasing trends and light to dark blue indicates significant increasing trends. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

accuracies in the range of 60% and 98% for greater and less than 50% lichen cover classes respectively. This suggests that the cover estimates developed here can be used to enhance existing land cover datasets with similar accuracies.

The regional analysis revealed similar performance between the north-west and north-central regions. The best results were achieved in the north-central region where the overall greatest quantity of lichen cover was observed, but these results were also related to reduced topographic effects. The north-east region was the least accurate due to several factors including: (1) lichen cover percentages were the lowest among the three regions, typically occurring in smaller, more discontinuous patches; (2) background, non-lichen cover (e.g. rock and soil) tended to be of a higher percentage and acted to reduce contrast with lichen as compared to the lichen/vegetation assemblages typical of the other regions; and (3) Landsat data was of reduced quality, containing more residual clouds, shadows, snow and topographic effects than the west and central regions. A confounding factor here was that the north-east region had the least amount of training/test data. It is difficult to speculate as to how additional data would improve results; however, future efforts will seek to increase field sampling particularly in the east. Differences observed between the global and regional modelling results may be reduced with more extensive training data in each region. The better results observed with the global model was considered to be due to the greater variation captured in the sample that improved training of the deep neural network.

We found the NN to be the best modelling approach in terms of accuracy and it did not require locally specific training. However, there are numerous modifications that could be explored to potentially improve results such as the network depth or width, activation functions, training strategies, etc. An important advantage of NNs compared to other methods are that they can be trained from existing data and later refined as new data becomes available. This is known as a transfer learning approach which is widely applied in deep learning (LeCun et al., 2015). NNs can also be optimized for local regions where more precise cover estimates are required. Convolution neural networks may provide further improvements; although, for sub-pixel estimation at moderate spatial resolutions, the contribution of spatial information in various convolution neural network architectures during initial experimentation did not reveal a significant benefit. Certain data transformations may also improve results. Here we tested using aspect directly or breaking it into directional components by applying sin and cos transformations. Results did not differ and thus aspect was used directly. However, for other network configurations with fewer layers or different activations functions this may be different, particularly where directional information is important and independent in the model.

4.2. Reference data insights

In this study, we sought to utilize an extensive collection of archived field data that represented a significant investment in data collection, particularly for remote locations. In most cases the lichen cover observations were based on visual interpretations made in situ (e.g. from point intercept sampling frames) or from field photographs. We recognize that such a diverse dataset suffers from subjectivity between interpreters and differences in sampling designs and data collection protocols can amount to differences in cover estimates. However, these data provided the opportunity to investigate the performance of our modelling approach across a large spatial extent – one not previously undertaken. Further, this large spatial extent allowed us to investigate regional differences in model performance.

Plot data can provide precise estimates of lichen cover, particularly for low-cover conditions. It also allows for understory assessment but often suffers from geolocation and limited scaling capacity. Scaling plot data is sometimes not possible because of insufficient ground coverage or the density of plots is not suitable to represent the sensor footprint (i.e. representative of pixel values found within a scene). Further, plot data may not align with the sensor footprint/pixel spacing. High

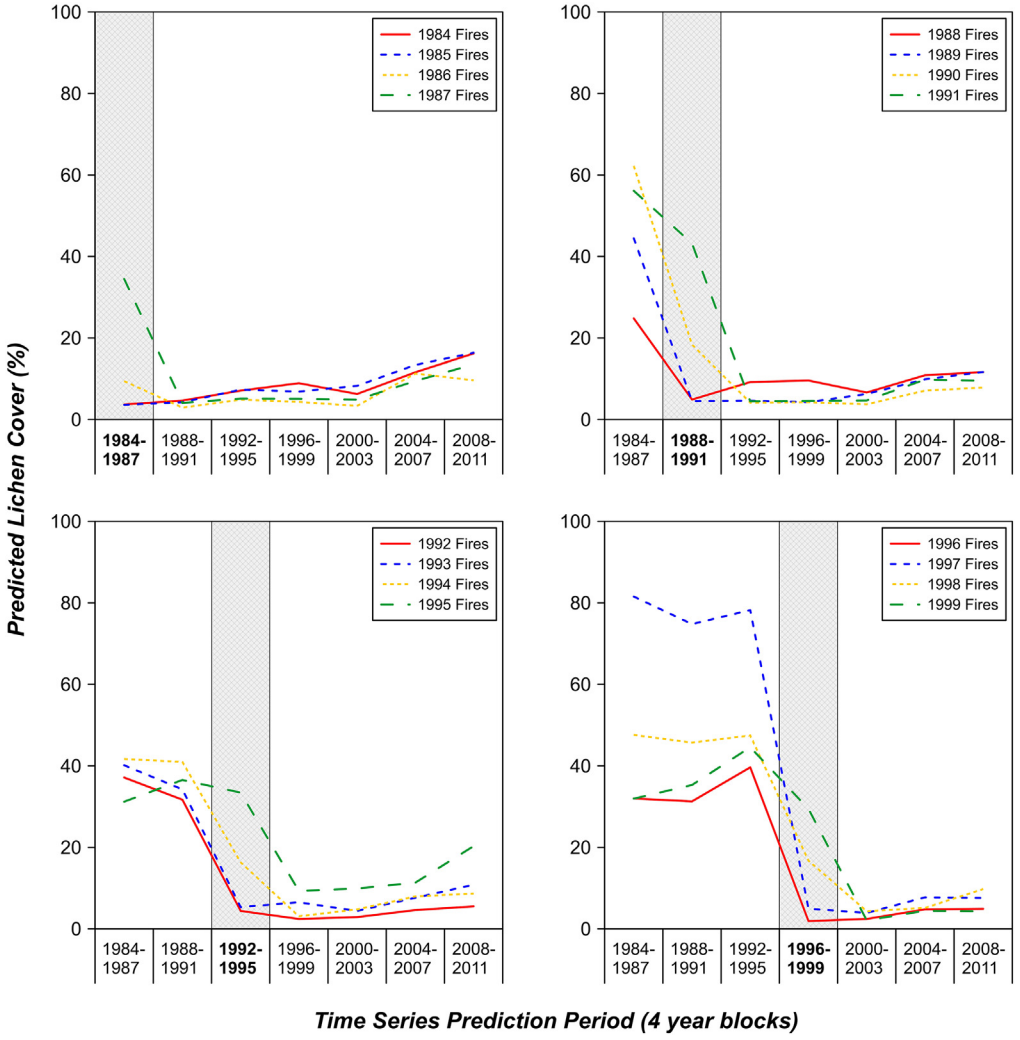


Fig. 8. Mean lichen cover predictions for different wildfires occurring within the north-central region (the coincident fire and Landsat period shown are in grey, zonal statistics for fire years are represented as separate colours/patterns, each panel represents a four-year grouping of forest fires).

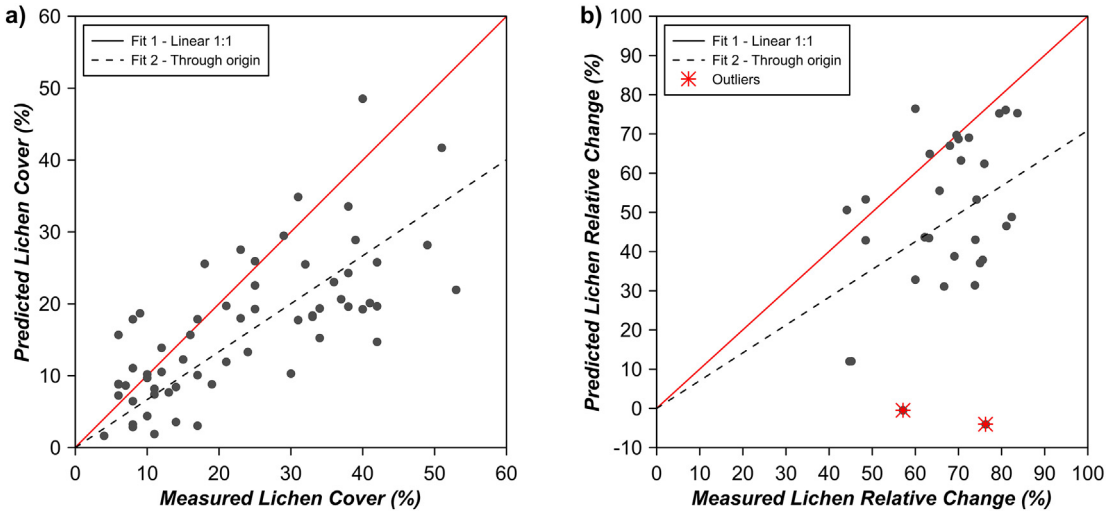


Fig. 9. Comparison of results with Fraser et al., 2014b for (a) predicted lichen percent cover and (b) relative change. Measured cover refers to Fraser et al., 2014b. The red asterisks in (b) represent samples contaminated by residual cloud and haze in the time series data. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

resolution air-photos can be effective but depending on resolution may limit the detectability of sparse cover lichen patches, especially those occurring below the canopy and where supporting ground data in not available. UAV's can provide scalable data acquired at high enough spatial resolutions for detection of lower discontinuous lichen cover levels if the vegetation canopy is not too dense (Fraser et al., 2016). Thus, a combination of UAV and plot sampling would be an effective approach for acquiring additional data.

4.3. Change assessment

Assessment for change detection showed that despite the model bias and random error components, detected changes were directionally consistent with known landscape change drivers in the regions tested (e.g. fire, greening, and ungulate activity). In this analysis, change was detected using the time series trend. Trend detection is a function of the length of the record, the size of the change, and the error in the observations (Pouliot et al., 2011; Saulquin et al., 2013). For a trend of 1% per year to be detected with an observation error rate of 10%, > 10 years of observations would be required. The change signal would not otherwise be distinguishable from the noise if the noise is randomly distributed in the temporal domain. In this analysis, the data spanned 28 years and suggests the potential to identify a change signal of < 0.5% per year (10% per 28 years). However, there are other factors to consider , such as the time series specific data errors, the trend estimator, and the time series observation density.

The errors observed in the time series were often related to residual cloud and cloud shadows in the Landsat composites. Improved screening is expected to enhance the results. The presence of haze was also found to influence results where spectral characteristics were indicative of lichen presence. In this case, haze inflated the lichen cover estimates. In future developments, reduction of visible bands in the feature set will be sought to reduce haze effects throughout the time series. A haze correction method such as that of Sun et al. (2017) should be examined.

The results of this study contribute to knowledge of baseline performance for comparison of time series methods and feature sets. The use of different trend methods than those used here may improve the results. The least absolute deviation regression may be more sensitive to change trends while being resistant to outliers. In our analysis, the low density of the time series and robust trend detection method led to instances of reduced sensitivity. Increasing the time series temporal density could be considered to increase sensitivity. In our analysis, cloud and shadow contamination was reduced by compositing Landsat observations in intervals of four years thus producing seven observations per pixel. Such an approach potentially excludes useful observations. Instead, a rolling compositing approach (e.g. a three-year moving window) could be used to generate annual estimates for almost all years within the Landsat archive. This would improve performance (i.e. detectability of slow change events) depending on the level of contamination in the time series. However, this approach will substantially increase processing time, and for many regions do not lead to improved results as there were insufficient clear-sky observations, such as in the north-east region.

5. Conclusion

We evaluated machine learning methods to assess broad-scale lichen cover for caribou in northern Canada using Landsat imagery, topographic and climate data. Our results were similar to previous regional studies with errors in the range of ± 10%. Of the models tested, a globally trained neural network showed the best performance and has the advantage of retraining with new data to improve results globally or specifically for local regions. The motivation to use Landsat 5 data was to allow for historical lichen cover change analysis. For study areas in the north-east, north-central, and north-west regions of Canada, change detection was consistent with known landscape change drivers associated with wildfire, caribou related foraging and trampling, and climate-induced shrub expansion. Although continued research is needed to improve model predictions, remote sensing-based methods are developing rapidly and will remain the best methods to monitor landscape-scale changes in lichen cover over remote regions of the world. A better understanding of factors affecting lichen abundance for caribou, and lichen recovery at a time of rapid climate change is critical to caribou and biodiversity conservation.

Author declaration

Blair Kennedy: Conceptualization, Methodology, Software, Validation, Investigation, Data curation, Writing - Original draft, Writing - Review & editing, Validation. **Darren Pouliot:** Conceptualization, Methodology, Software, Validation, Formal analysis, Data curation, Writing - Original draft, Writing - Review & editing, Validation, Supervision. **Micheline Manseau:** Conceptualization, Data curation, Writing - Original draft, Writing - Review & editing. **Robert Fraser:** Data curation, Writing - Original draft, Writing - Review & editing. **Jason Duffe:** Conceptualization, Funding acquisition, Project administration, Writing - Review & editing. **Jon Pasher:** Funding acquisition, Project administration, Writing - Review & editing. **Wenjun Chen:** Data curation, Writing - Review & editing. **Ian Olthof:** Data curation, Writing - Review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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